

Customer Shopping Behavior Analysis

Uncovering insights from 3,900 purchases to drive strategic business decisions



Dataset at a Glance

3,900

Total Purchases

Transactions analyzed

18

Data Points

Features per transaction

50

Locations

Geographic coverage

25

Products

Items tracked

Demographics

Age, gender, location, subscription status

Purchase Details

Category, amount, season, size, color



Data Preparation Journey

01

Data Loading

Imported dataset using pandas

02

Initial Exploration

Structure check and summary statistics

03

Missing Data

Imputed 37 missing review ratings by category median

04

Standardization

Renamed columns to snake_case

05

Feature Engineering

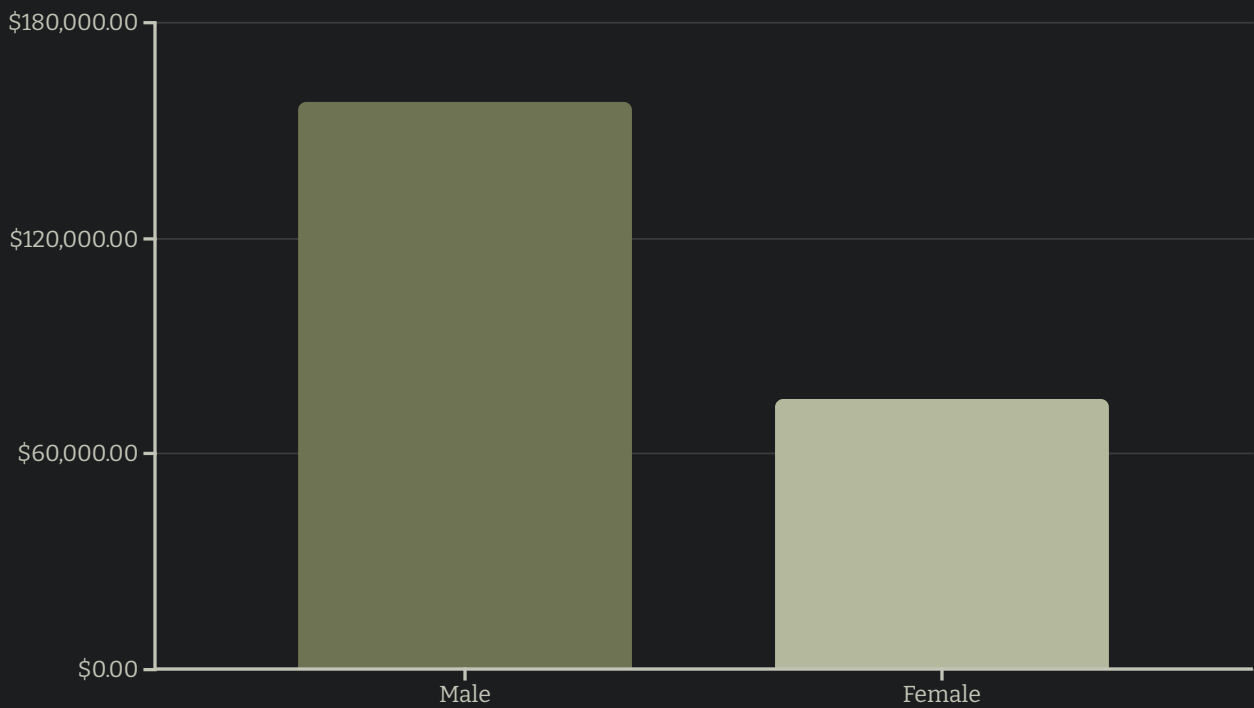
Created age_group and purchase_frequency_days

06

Database Integration

Loaded into PostgreSQL for SQL analysis

Revenue Insights



Gender Revenue Gap

Male customers generate 2.1x more revenue than female customers

Total revenue: \$233,081

Discount Strategy Analysis



High-Spending Discount Users

839 customers used discounts but spent above average



Top Discount Products

- Hat: 50% discount rate
- Sneakers: 49.66%
- Coat: 49.07%

Balance sales boosts with margin control—discounts drive volume but may impact profitability



Product Performance Leaders

Top Rated Products

1. Gloves: 3.86 rating
2. Sandals: 3.84
3. Boots: 3.82
4. Hat: 3.80
5. Skirt: 3.78

Best Sellers by Category

Accessories: Jewelry (171 orders)

Clothing: Blouse & Pants (171 each)

Footwear: Sandals (160 orders)

Outerwear: Jacket (163 orders)



Customer Segmentation



Strong loyal base presents opportunity for retention programs and upselling

Subscription & Shipping Patterns

Subscription Status

Subscribers: 1,053 (avg spend \$59.49)

Non-subscribers: 2,847 (avg spend \$59.87)

Minimal spending difference suggests untapped subscription value

Shipping Preferences

Express: \$60.48 avg purchase

Standard: \$58.46 avg purchase

Express users spend 3.5% more per transaction

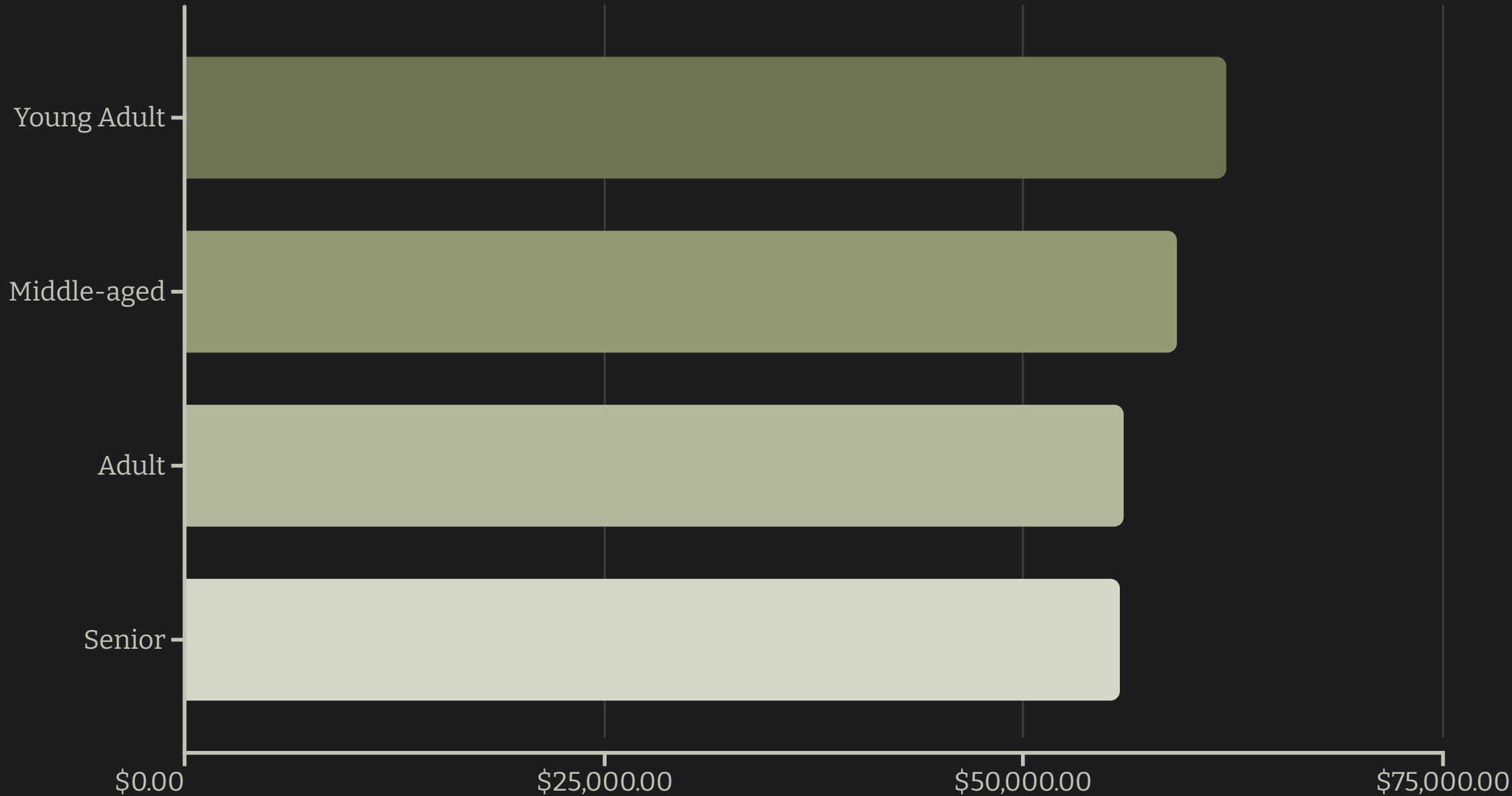
Repeat Buyer Behavior

Customers with 5+ purchases:
3,476 total

Only 958 are subscribers—
huge conversion opportunity



Revenue by Age Demographics



Young adults lead revenue generation—target marketing campaigns to this demographic

Strategic Recommendations



Boost Subscriptions

Promote exclusive benefits to convert 2,518 repeat buyers into subscribers



Loyalty Programs

Reward repeat buyers to strengthen the 79.9% loyal customer base



Review Discount Policy

Balance promotional impact with margin protection on high-discount items



Product Positioning

Highlight top-rated products (Gloves, Sandals, Boots) in marketing campaigns



Targeted Marketing

Focus on young adults and express-shipping users for maximum ROI